Chronic Kidney Disease (CKD) Prediction

This is ML, Supervised, Classification problem as the Client has given Output, which is CKD disease prediction and has a very clear requirement.

The given dataset has 399 rows, 24 input columns - 12 numerical and 12 categorical data, in which 1 is output column.

Since there are 12 Categorical data columns and they are Nominal data. So, we use getdummies to convert to number. Now the columns has increased to 15.
Total columns now are 28, out of which 1 is output column and 27 are input columns.

Used **12 Classification models** and used ROC_AUC and F1 Score for evaluation.

	roc_auc	f1_score	best_params
GaussianNB	1.000000	1.000000	{}
SVC	1.000000	0.979951	{'C': 10, 'gamma': 'auto', 'kernel': 'poly'}
CatBoostClassifier	1.000000	0.979951	{'depth': 3, 'iterations': 50, 'learning_rate'...
AdaBoostClassifier	1.000000	1.000000	{'learning_rate': 0.1, 'n_estimators': 50}
BernoulliNB	0.999596	0.980032	{}
LGBMClassifier	0.999596	0.979951	{'boosting_type': 'gbdt', 'learning_rate': 0.1...
XGBClassifier	0.999596	0.969881	{'learning_rate': 0.1, 'max_depth': 3, 'n_esti...
LogisticRegression	0.998384	0.960000	{'C': 0.1, 'l1_ratio': 0.5, 'max_iter': 1000, ...
RandomForestClassifier	0.998182	0.979951	{'criterion': 'gini', 'max_features': 'sqrt', ...
KNeighborsClassifier	0.990505	0.990009	{'algorithm': 'auto', 'n_neighbors': 3, 'weigh...
MultinomialNB	0.964444	0.870273	{}
DecisionTreeClassifier	0.957576	0.959903	{'criterion': 'gini', 'max_features': 'sqrt', ...

This shows Guassian Naive Bayes and AdaBoost Classifier as the best models, because the ROC_AUC and F1_Score values are 1, which is the highest.

Lets see Confusion matrix and Classification Report for these models.

<table>
<tr>
<td>Guassian Naive Bayes model</td>
<td><pre>
Best Parameters: {}
F1 Score (weighted): 1.0000
Confusion Matrix:
 [[45 0]
 [0 55]]
Classification Report:
               precision   recall f1-score   support

           0   1.00   1.00   1.00   45
           1   1.00   1.00   1.00   55

    accuracy   1.00   100
   macro avg   1.00   1.00   1.00   100
weighted avg   1.00   1.00   1.00   100

ROC AUC Score: 1.0000
</pre></td>
</tr>
<tr>
<td>AdaBoost Classifier model</td>
<td><pre>
Best Parameters: {'learning_rate': 0.1, 'n_estimators': 50}
F1 Score (weighted): 1.0000
Confusion Matrix:
 [[45 0]
 [0 55]]
Classification Report:
               precision   recall f1-score   support

           0   1.00   1.00   1.00   45
           1   1.00   1.00   1.00   55

    accuracy   1.00   100
   macro avg   1.00   1.00   1.00   100
weighted avg   1.00   1.00   1.00   100

ROC AUC Score: 1.0000
</pre></td>
</tr>
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So, based on sorting result, Best Model used: GaussianNB with ROC AUC = 1.0000 and F1 Score = 1.0000